



AtliQ Hardwares



CONSUMER GOODS DOMAIN

SUPPLY CHAIN ANALYSIS

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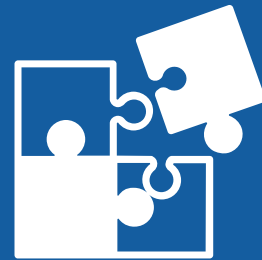
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ABOUT US



ATLIQ – INTRO

Atliq Hardwares (imaginary company) is one of the leading computer hardware producers in India and well expanded in other countries too.



BACKGROUND / PROBLEM

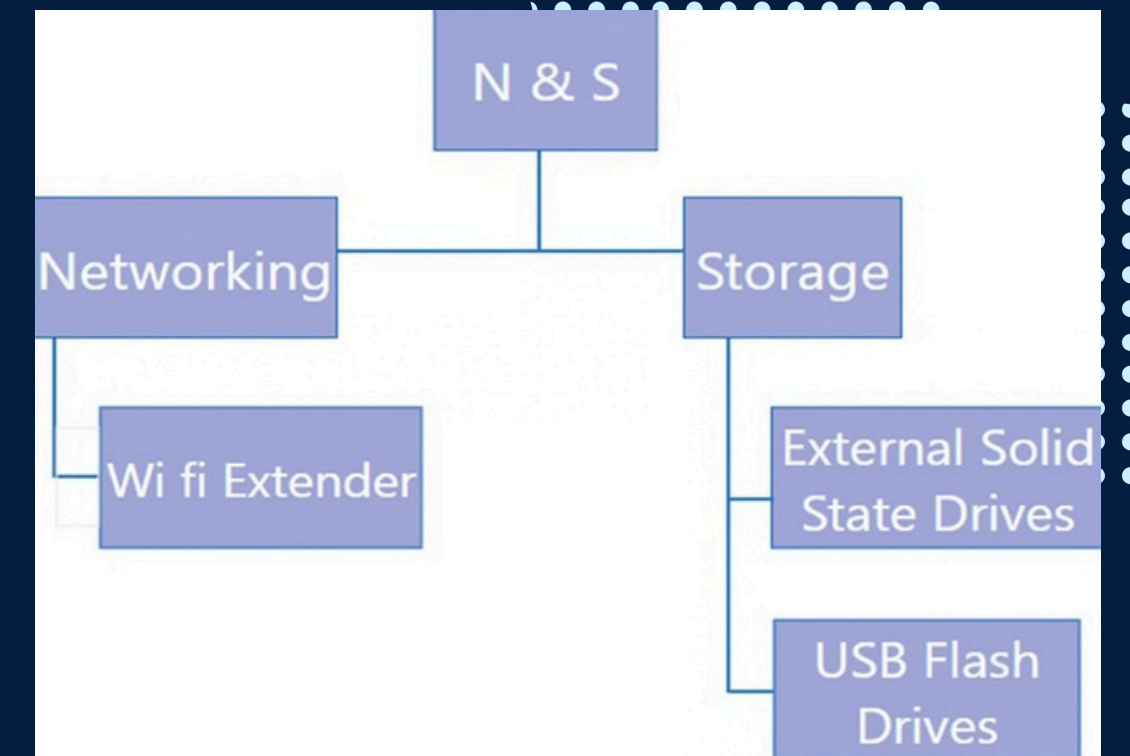
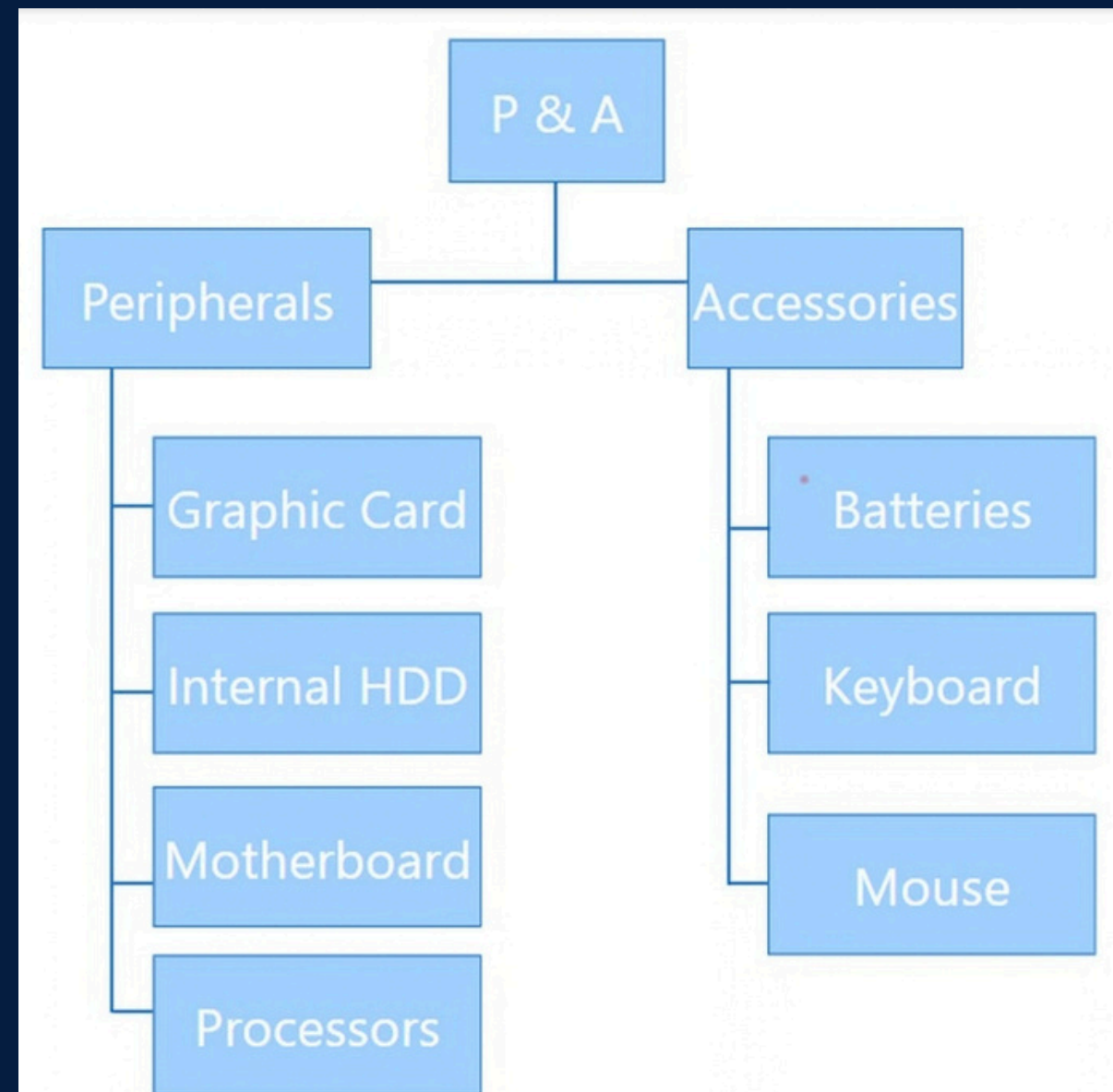
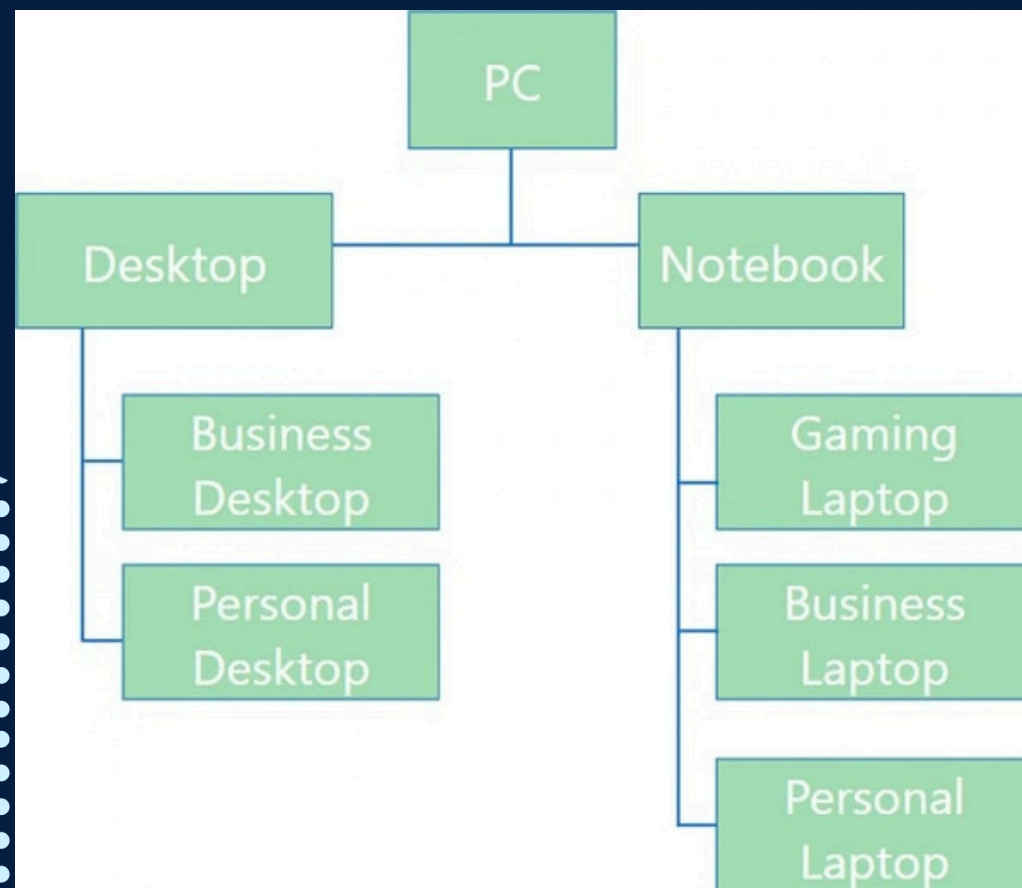
The management noticed that they do not get enough insights to make quick and smart data-informed decisions.



ATLIQ'S MARKET



ATLIQ'S PRODUCT LINE



dim_customer

dim_product

fact_gross_price

fact_manufacturing_
cost

fact_pre_invoice_
deductions

fact_sales_
monthly

DATA





Codebasics SQL Challenge

Requests:

1. Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.
2. What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields,
unique_products_2020
unique_products_2021
percentage_chg
3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields,
segment
product_count
4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields,
segment
product_count_2020
product_count_2021
difference
5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields,
product_code
product
manufacturing_cost

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AD-HOC INSIGHTS



6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market. The final output contains these fields,
customer_code
customer
average_discount_percentage
7. Get the complete report of the Gross sales amount for the customer **"Atliq Exclusive"** for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.
The final report contains these columns:
Month
Year
Gross sales Amount
8. In which quarter of 2020, got the maximum total_sold_quantity? The final output contains these fields sorted by the total_sold_quantity,
Quarter
total_sold_quantity
9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields,
channel
gross_sales_mln
percentage
10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021? The final output contains these fields,
division
product_code

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Ad-hoc requests, queried results and visualizations

Request 1: Which markets does customer "Atliq Exclusive" operate in within the APAC region?

Market
India
Indonesia
Japan
Philippines
South Korea
Australia
New Zealand
Bangladesh

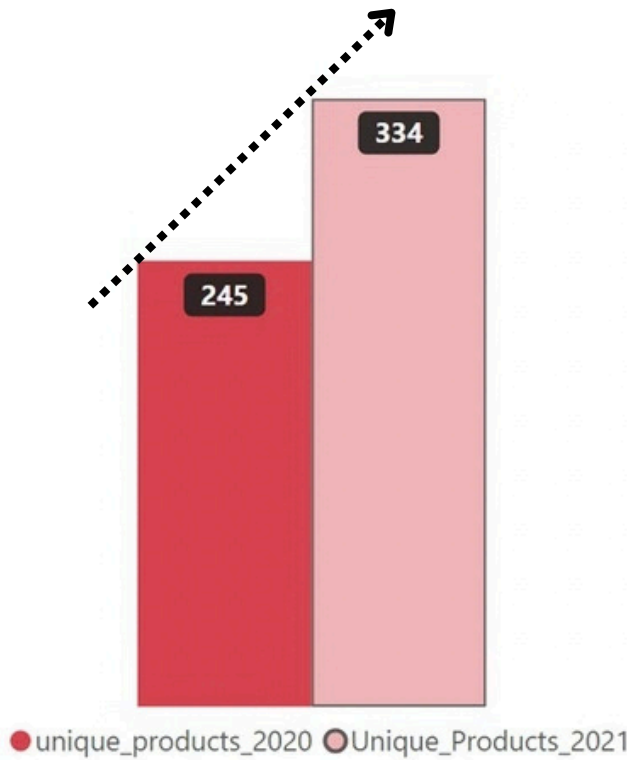


REQUEST 2: WHAT IS THE PERCENTAGE OF UNIQUE PRODUCT INCREASE IN 2021 VS. 2020?

unique_products_2020	unique_products_2021	percentage_chg
245	334	36.33%

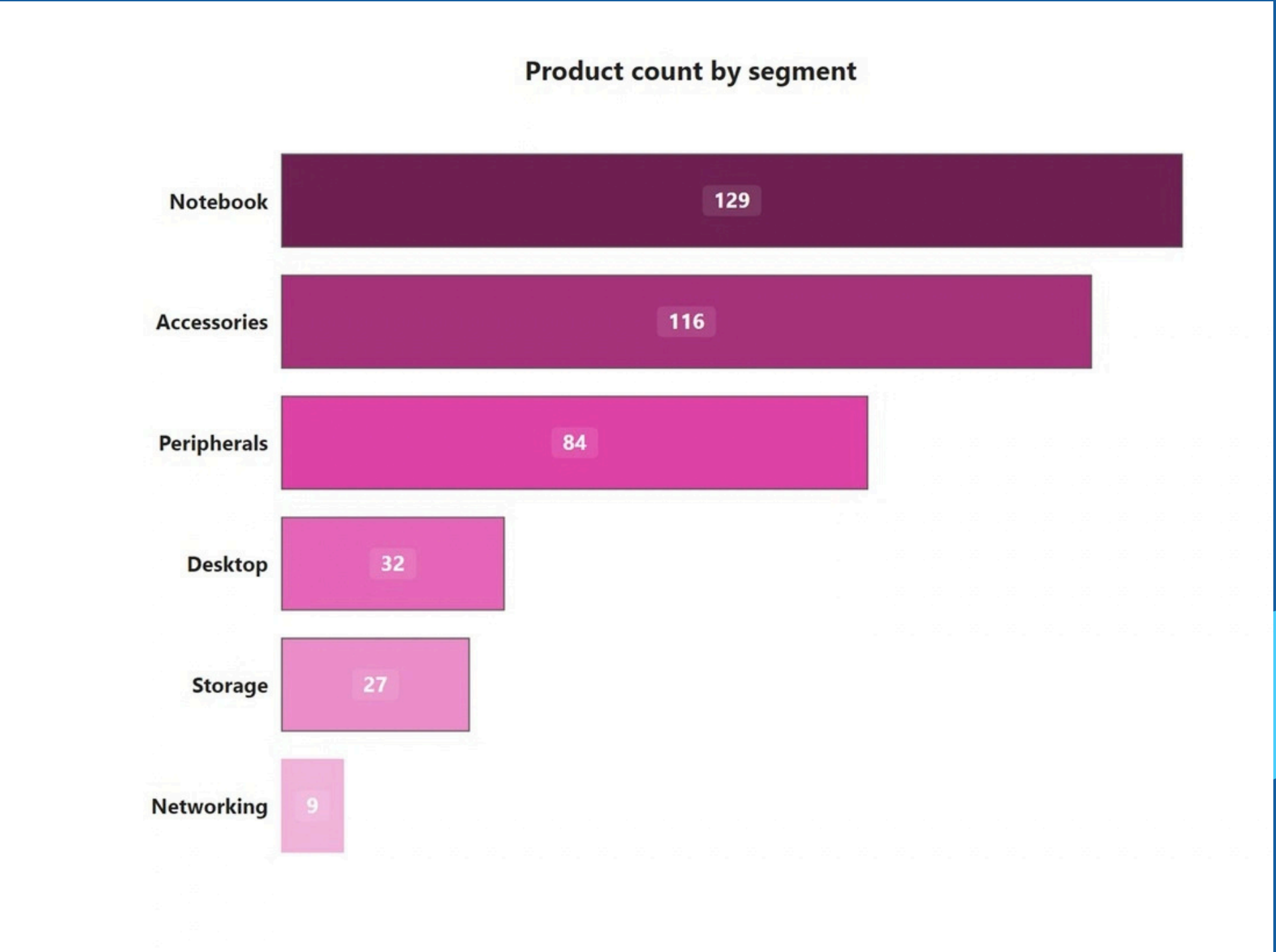


33.6% 



REQUEST 3: PROVIDE A REPORT WITH ALL THE UNIQUE PRODUCT COUNTS FOR EACH SEGMENT AND SORT THEM IN DESCENDING ORDER OF PRODUCT COUNTS.

Segment	Product Count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

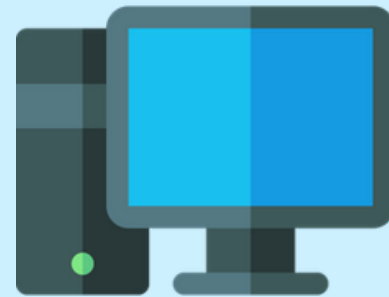


REQUEST 4: WHICH SEGMENT HAD THE MOST INCREASE IN UNIQUE PRODUCTS IN 2021 VS 2020?

Segment	product_count_2020	product_count_2021	Difference
Accessories	6	9	3
Notebook	7	22	15
Peripherals	12	17	5
Desktop	59	75	16
Storage	69	103	34
Networking	92	108	16

Segment	Product Count 2020	Product Count 2021	Difference
Accessories	69	103	34 ↑
Notebook	92	108	16 ↑
Peripherals	59	75	16 ↑
Desktop	7	22	15 ↑
Storage	12	17	5 ↑
Networking	6	9	3 ↑
Total	245	334	89

REQUEST 5: GET THE PRODUCTS THAT HAVE THE HIGHEST AND LOWEST MANUFACTURING COSTS.



AQ HOME Allin1
Gen 2

Personal Desktop

\$240.54

A6120110206

highest manufacturing product



AQ Master wired
x1Ms
Mouse

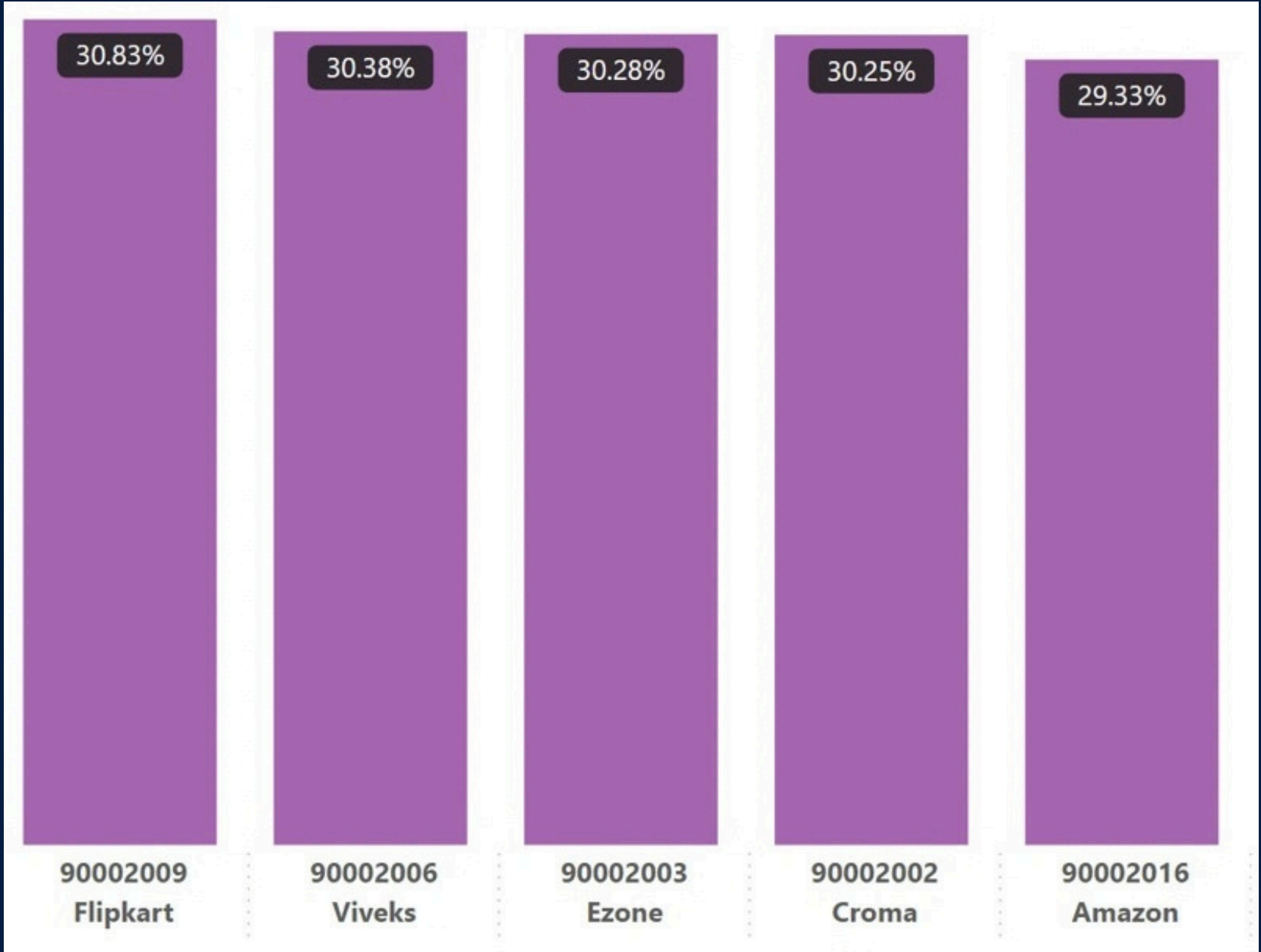
\$0.89

A2118150101

lowest manufacturing product

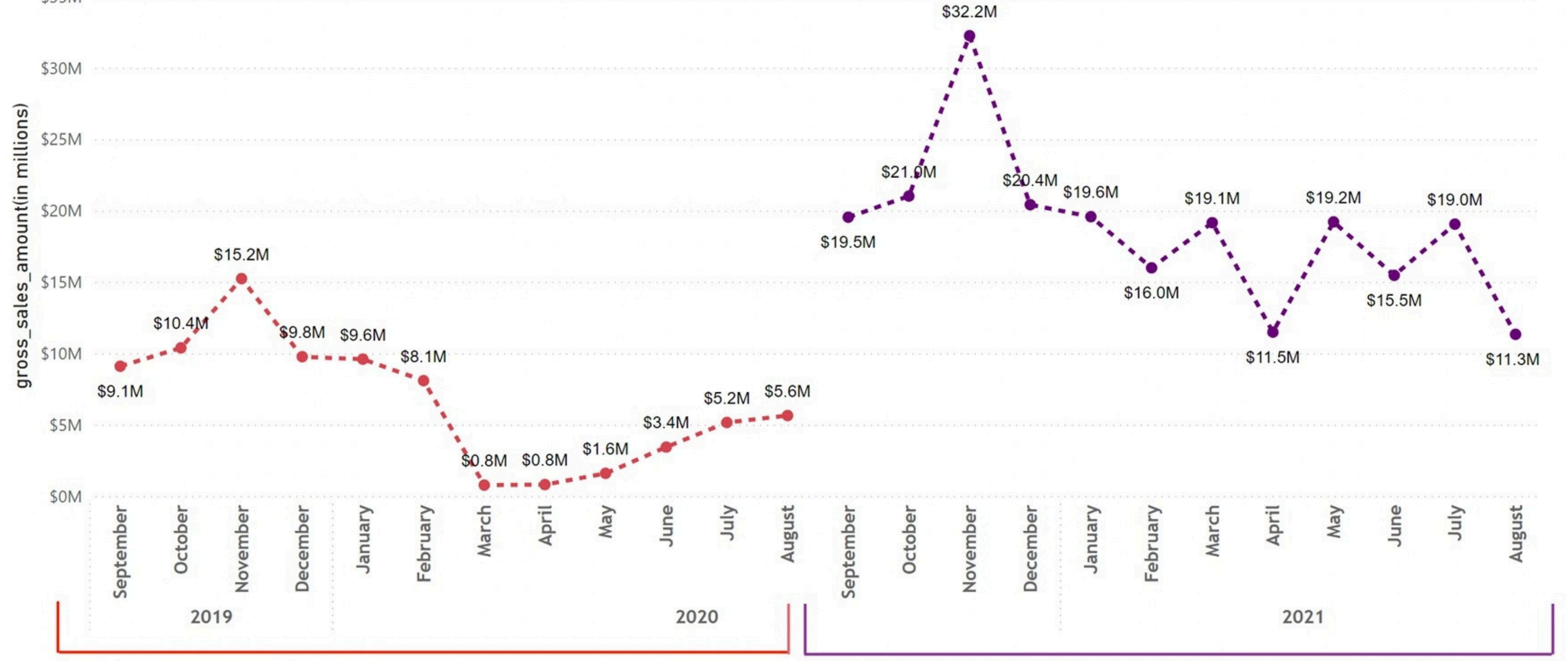
REQUEST 6: GENERATE A REPORT WHICH CONTAINS THE TOP 5 CUSTOMERS WHO RECEIVED AN AVERAGE HIGH PRE_INVOICE_DISCOUNT_PCT FOR THE FISCAL YEAR 2021 AND IN THE INDIAN MARKET.

customer_code	customer	average_discount_percentage
90002009	Flipkart	30.83%
90002006	Viveks	30.38%
90002003	Ezone	30.28%
90002002	Croma	30.25%
90002016	Amazon	29.33%



Month	fiscal_year	gross_sales_amount
September (2019)	2020	9.09M
October (2019)	2020	10.38M
November (2019)	2020	15.23M
December (2019)	2020	9.76M
January (2020)	2020	9.58M
February (2020)	2020	8.08M
March (2020)	2020	0.77M
April (2020)	2020	0.80M
May (2020)	2020	1.59M
June (2020)	2020	3.43M
July (2020)	2020	5.15M
August (2020)	2020	5.64M
September (2020)	2021	19.53M
October (2020)	2021	21.02M
November (2020)	2021	32.25M
December (2020)	2021	20.41M

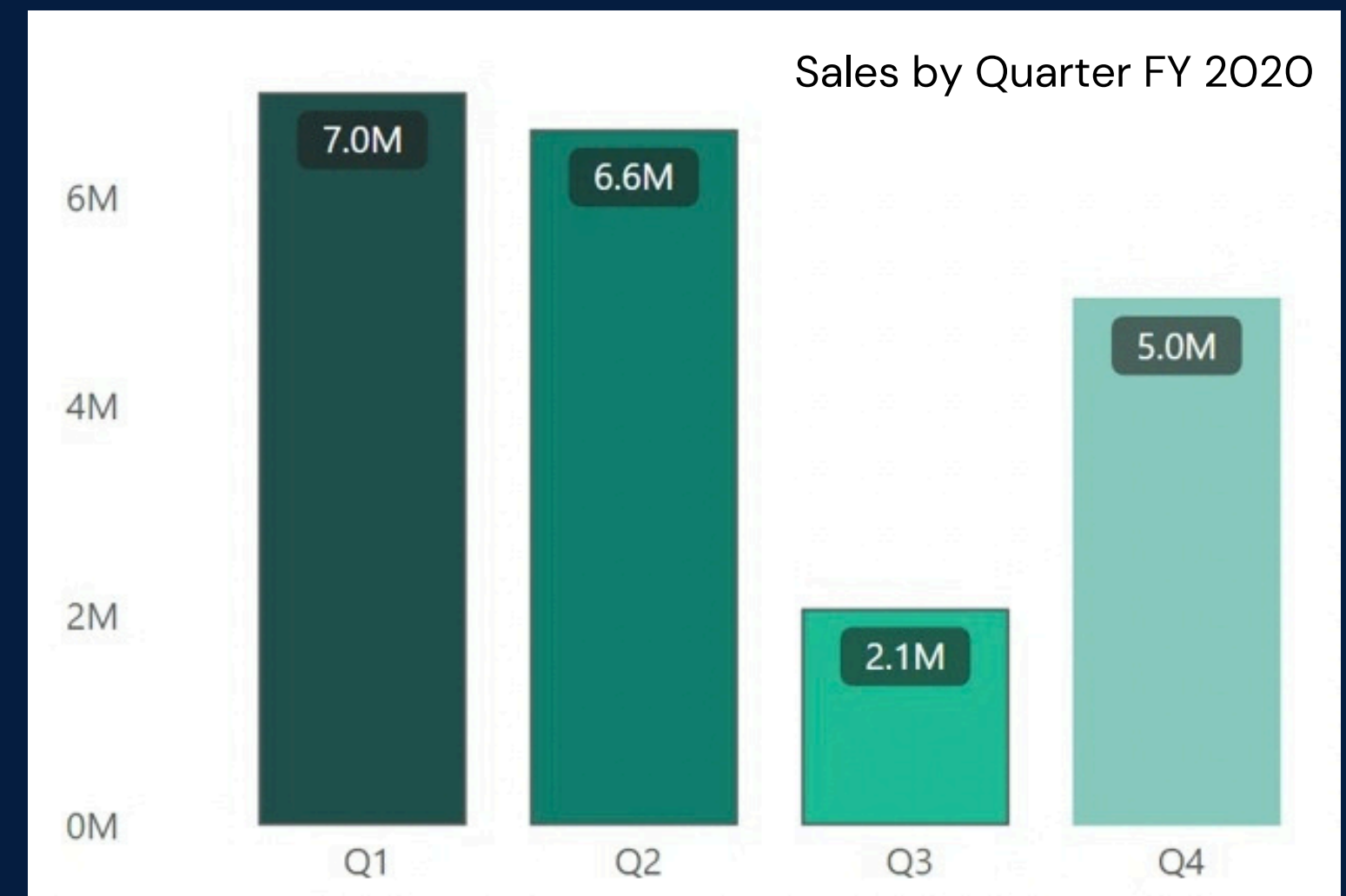
Request 7: Get the complete report of the Gross sales amount for the customer "Atliq Exclusive" for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.



Sales Trend - AtliQ Exclusive
Fiscal Year: September to August

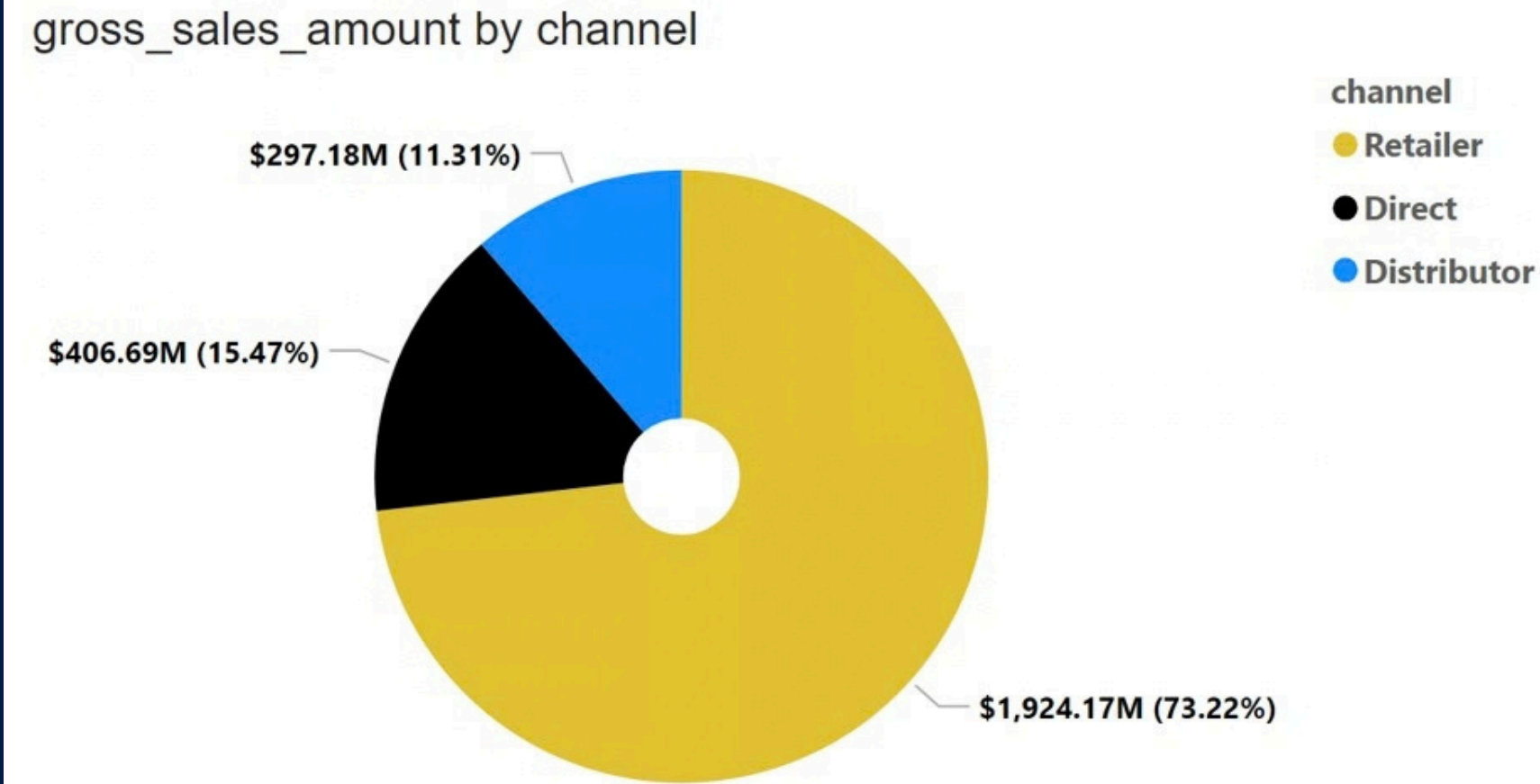
REQUEST 8: IN WHICH QUARTER OF 2020, GOT THE MAXIMUM TOTAL_SOLD_QUANTITY?

Quarter	total_sold_quantity
Q1	7005619
Q2	6649642
Q4	5042541
Q3	2075087



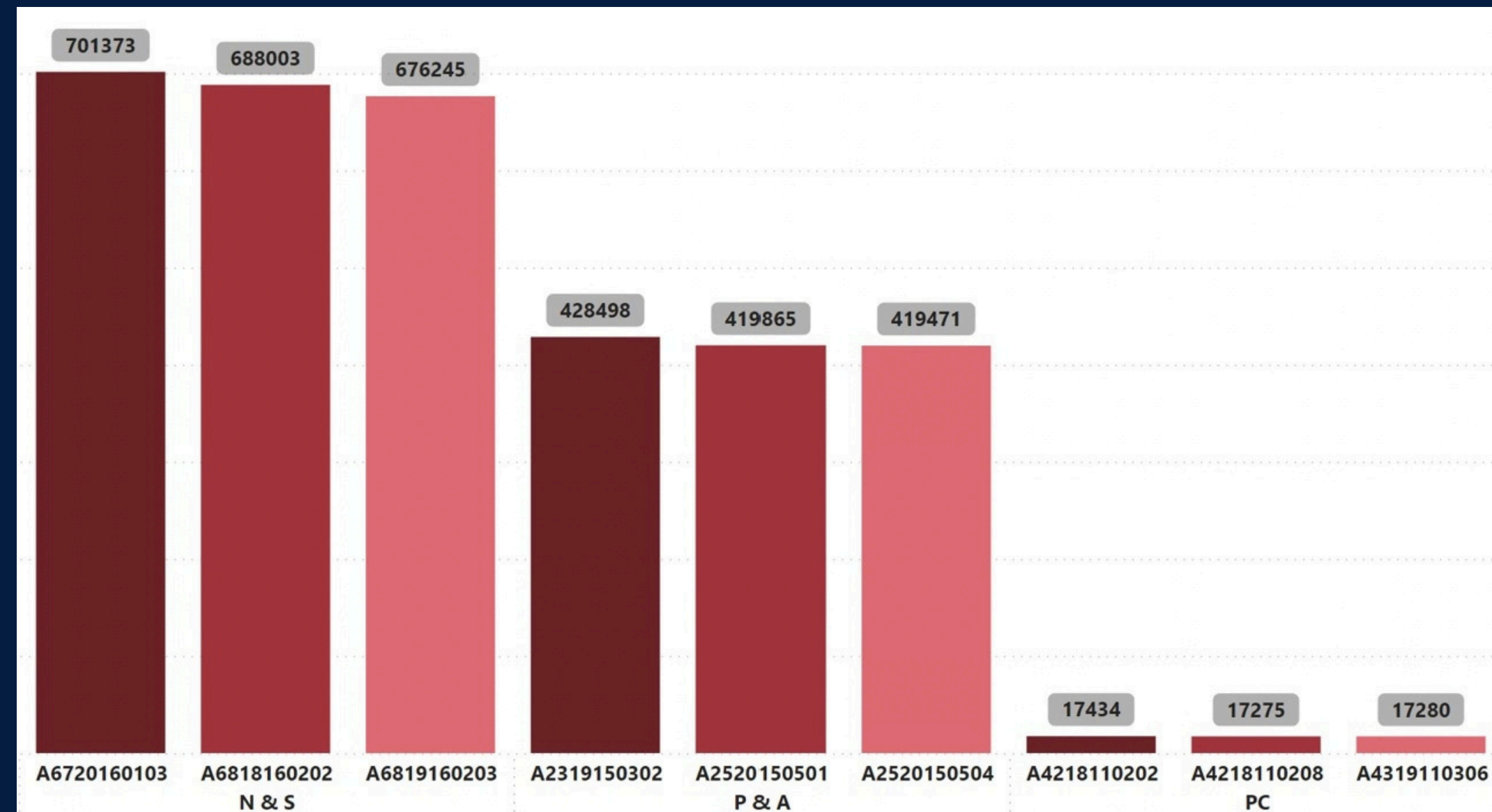
Request 9: Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

Channel	gross_sales_mln	percentage
Retailer	\$ 1924.17	73.22%
Direct	\$ 406.69	15.48%
Distributor	\$ 297.18	11.31%



REQUEST 10: GET THE TOP 3 PRODUCTS IN EACH DIVISION THAT HAVE A HIGH TOTAL_SOLD_QUANTITY IN THE FISCAL_YEAR 2021?

Division	product_code	product	total_sold_quantity	rank_order
N & S	A6720160103	Pen Drive 2 IN 1	701373	1
N & S	A6818160202	Pen Drive DRC	688003	2
N & S	A6819160203	Pen Drive DRC	676245	3
P & A	A2319150302	Gamers Ms	428498	1
P & A	A2520150501	Maxima Ms	419865	2
P & A	A2520150504	Maxima Ms	419471	3
PC	A4218110202	Digit	17434	1
PC	A4319110306	Velocity	17280	2
PC	A4218110208	Digit	17275	3



INSIGHTS

01

The unique products saw an impressive 36.33% increase in sales during the 2021 fiscal year compared to the previous year.

02

The Notebook emerged as the top-selling product, while the Networking product line recorded the lowest sales figures.

03

In terms of manufacturing costs, Desktops had the highest expenditure, while Mouse production costs were the lowest.

04

For FY2021, Flipkart made the largest customer contribution, accounting for a notable 30.83%, whereas Amazon had the lowest at 29.33%.

05

March 2020 marked the lowest sales period, with 2.8 million units sold, while sales peaked in November 2021, reaching a record 32.2 million units.